COMPUTER NETWORKS PROJECT

AI-Based Real-Time DDoS Detection & Mitigation System

Observe traffic. Detect anomalies. Defend automatically.

INTERACTIVE PROTOTYPE | REAL-TIME ANALYTICS | AUTOMATED RESPONSE

PROJECT TEAM

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AI-POWERED NETWORK DEFENSE COMMAND CENTER

Executive Snapshot

A complete detection-to-response workflow in one security command center.

1 OBSERVE

Continuously visualize packet velocity, bandwidth pressure, IP entropy, protocol mix, and deviation from a rolling baseline.

2 DECIDE

Combine multiple signals into a threat-confidence score instead of relying on one fixed traffic threshold.

3 DEFEND

Trigger a controlled firewall policy, drop malicious packets, and create an auditable blocked-threat record.

WHY THIS PROJECT MATTERS

DDoS attacks target **availability**. Our system reduces the gap between visibility, intelligent classification, and controlled defensive action.

WINDOW	REFRESH	PIPELINE
30 SEC	1 SEC	5 STAGES

Understanding the DDoS Threat

Distributed attacks overwhelm network, protocol, or application resources to deny service to legitimate users.

L3 / L4

VOLUMETRIC

Consumes upstream bandwidth

UDP flood • DNS amplification

L7

APPLICATION

Targets expensive operations

HTTP GET • API flood



L3 / L4

PROTOCOL

Exhausts connection state

SYN flood • ACK flood

HYBRID

MULTI-VECTOR

Combines changing techniques

Distributed + adaptive behavior

CORE CHALLENGE

Separate a malicious flood from a legitimate flash crowd—without blocking real users.

Project Objectives & Design Requirements

The system is designed for visibility, explainability, safe automation, and repeatable demonstration.

01

REAL-TIME VISIBILITY

Track packet rate, byte rate, protocol behavior, and source diversity in a live rolling window.

02

MULTI-FEATURE DETECTION

Combine request velocity, bandwidth, and IP entropy to reduce single-threshold false alarms.

03

CONTROLLED MITIGATION

Separate model inference from policy authorization before applying a firewall action.

04

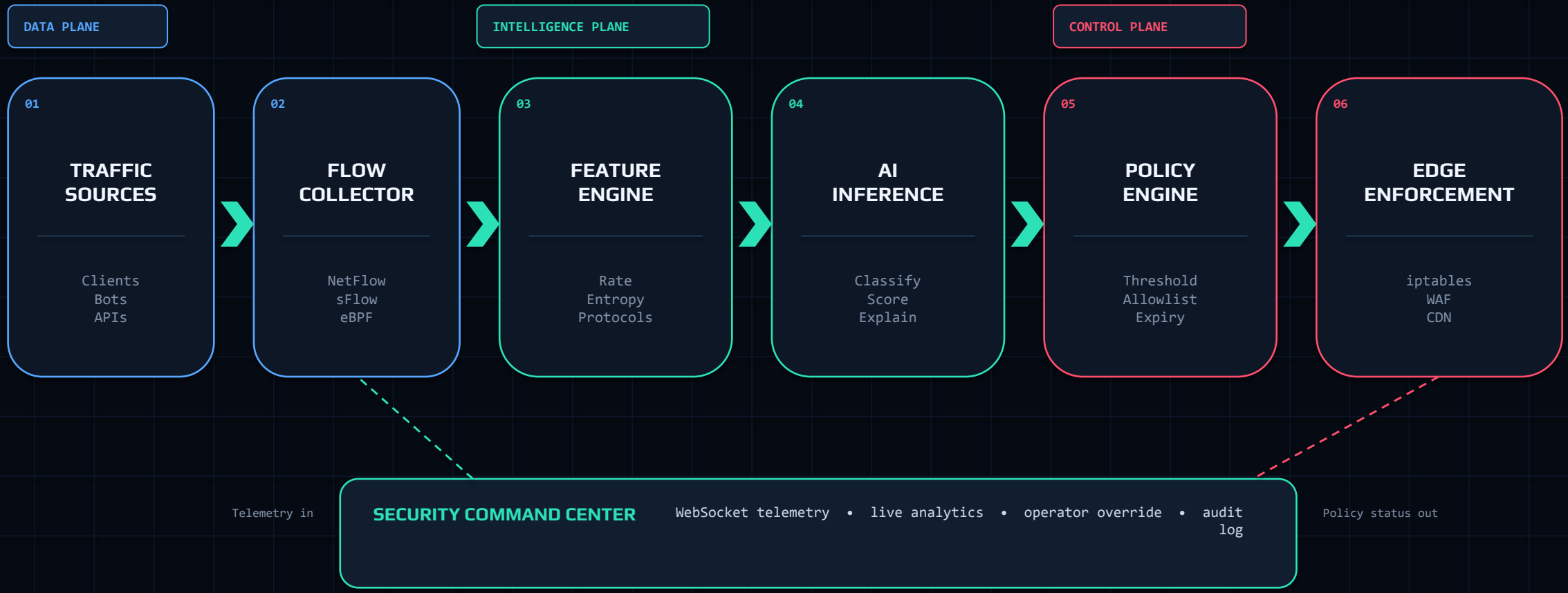
AUDITABLE RESPONSE

Record timestamp, source IP, attack category, action, and enforcement status for every event.

DESIGN PRINCIPLE: MODEL RECOMMENDS • POLICY AUTHORIZES • ENFORCEMENT ACTS

End-to-End System Architecture

A modular pipeline keeps traffic collection, AI inference, policy decisions, enforcement, and visualization independent.



Detection Intelligence

The decision combines several behavioral signals—high traffic alone is never treated as proof of an attack.

FEATURE VECTOR



MODEL STRATEGY

SUPERVISED	Random Forest • XGBoost
ANOMALY	Isolation Forest • Autoencoder

A hybrid approach combines explainable rules with adaptive inference.

EVALUATION THAT MATTERS

- Precision
- Recall
- F1 score
- False-positive rate
- Inference latency

Never claim “100% accurate.” Security performance must be measured on realistic, imbalanced traffic.

From Telemetry to a Trusted Decision

A gated workflow limits false positives and prevents an AI prediction from directly executing unrestricted commands.



✓

BENIGN / BELOW THRESHOLD

Continue monitoring • preserve user access

!

CONFIRMED / ABOVE THRESHOLD

Mitigate • log • expire rule • verify recovery

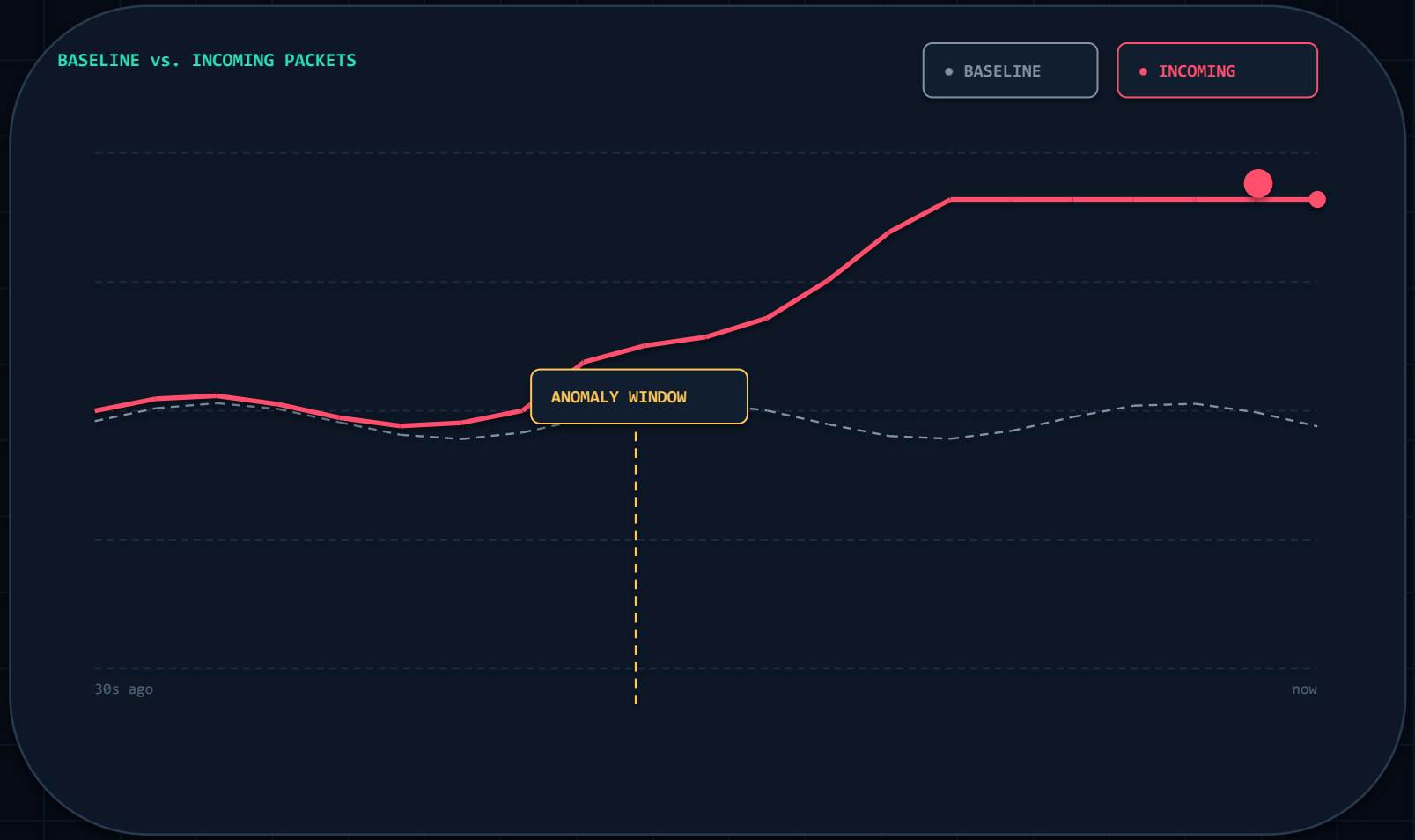
Interactive Security Command Center

A responsive browser dashboard transforms complex telemetry into one clear operational picture.



30-Second Rolling Traffic Analysis

Normal traffic follows the learned baseline; a sustained multi-feature surge creates a clear separation.



THREAT CONFIDENCE

99.4%

CRITICAL

1. **Sustained burst**
   Not a one-time spike
2. **Entropy divergence**
   Distributed sources
3. **Bandwidth pressure**
   Near link capacity
4. **Baseline deviation**
   Multi-signal agreement

Automated Mitigation — Fast, Controlled, Reversible

The response engine applies a restricted policy only after detection confidence and safety conditions are satisfied.

AUTO-MITIGATION ACTIVE Applying iptables firewall block rule. POLICY AUTHORIZED

ENFORCEMENT EXAMPLE

```
# Temporary host-level block
iptables -I INPUT -s 203.0.113.42 -j DROP

# Rate-limit suspicious new connections
iptables -A INPUT -p tcp --syn -m limit \
  --limit 25/second -j ACCEPT
```

Illustrative rule • execute only through a privileged policy agent

MITIGATION GUARDRAILS

01	Confidence threshold	Require multi-feature agreement
02	Progressive response	Rate-limit before hard block
03	Rule expiry	Automatically remove temporary rules
04	Operator override	Pause, approve, or roll back policy

DETECT → VALIDATE → AUTHORIZE → ENFORCE → VERIFY

Blocked Threat Log & Audit Trail

Every automated decision is visible, attributable, and reviewable by an operator.

BLOCKED IPS	DROPPED PACKETS	POLICY STATUS
24	1.82M	ACTIVE

TIMESTAMP	ATTACKER IP	ATTACK TYPE	ACTION TAKEN	STATUS
10:42:18	203.0.113.42	UDP Flood	iptables DROP rule applied	BLOCKED
10:42:19	198.51.100.77	SYN Flood	Edge rate limit + quarantine	MITIGATED
10:42:21	192.0.2.146	HTTP GET Flood	iptables DROP rule applied	BLOCKED
10:42:23	203.0.113.98	DNS Amplification	Upstream filter requested	MITIGATED

AUDIT PRINCIPLE

Record what was observed, why the policy triggered, what action ran, and whether service recovered.

Technology Stack & Deployment Path

The current showcase is lightweight and offline-ready; the architecture defines clear integration points for production telemetry.

CURRENT INTERACTIVE PROTOTYPE

Runs entirely in the browser

HTML5	Semantic command-center structure
CSS3	Responsive dark cyber UI + animation
JavaScript	State, simulation, chart, live logs
SVG	Gauges and rolling traffic visualization
Offline	No API key, backend, or internet required

Purpose: safely demonstrate the complete detection-to-response interaction.

PRODUCTION INTEGRATION PATH

Streaming, scalable, enforceable

COLLECT	NetFlow / sFlow / eBPF / proxy logs
STREAM	Kafka or secure event transport
INFER	Python model service + feature store
PUSH	WebSocket telemetry to dashboard
ENFORCE	Restricted policy agent at the edge

Principle: replace the simulator without redesigning the user interface.

Security, Ethics & Current Limitations

A strong engineering project states what is proven, what is simulated, and what must be validated next.

PROTOTYPE DISCLOSURE

This browser build uses controlled telemetry and simulated enforcement. It does not inspect campus traffic or modify the laptop firewall.

1 **SIMULATED TELEMETRY**

Production requires a real flow or packet-metadata collector.

2 **MODEL VALIDATION**

Training, cross-validation, drift testing, and realistic benign data are still required.

3 **FALSE POSITIVES**

Flash crowds can resemble attacks; progressive controls must protect real users.

4 **UPSTREAM CAPACITY**

Host rules cannot recover bandwidth already saturated before the server.

5 **PRIVILEGE BOUNDARY**

Only a restricted backend agent should be allowed to execute firewall policy.

6 **RULE LIFECYCLE**

Blocks need scope limits, expiry, rollback, allowlists, and operator review.

HONEST LIMITATIONS BUILD CREDIBILITY

Future Roadmap

A staged path from interactive concept to measurable, production-grade network defense.

PHASE 01

1

CONNECT

WebSocket telemetry API
NetFlow/eBPF collector
Secure event schema

PHASE 02

2

TRAIN

CIC-DDoS2019 baseline
Local benign traffic
Model comparison

PHASE 03

3

HARDEN

Signed policy requests
Rule expiry + rollback
Role-based access

PHASE 04

4

VALIDATE

Controlled load tests
Precision / recall / F1
End-to-end latency

PHASE 05

5

SCALE

Multi-edge deployment
WAF/CDN integration
Drift monitoring

1 2 3 4 5

CONCEPT

PRODUCTION-READY

PROJECT TEAM

Built as one integrated cybersecurity workflow.

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OBSERVABLE INTELLIGENT AUTOMATED ACCOUNTABLE

AI-Based Real-Time DDoS Detection & Mitigation



THANK YOU

Security is not a single alert.
It is a controlled response loop.

Technical References & Validation Sources

Standards, public datasets, and documentation that support the production implementation path.

01

CIC-DDoS2019 Dataset

Canadian Institute for Cybersecurity, University of New Brunswick

unb.ca/cic/datasets/ddos-2019.html

02

MITRE ATT&CK — T1498

Network Denial of Service technique and mitigations

attack.mitre.org/techniques/T1498

03

NIST Cybersecurity Framework 2.0

Govern, identify, protect, detect, respond, and recover

nist.gov/cyberframework

04

Netfilter / iptables Documentation

Linux packet filtering and policy enforcement reference

netfilter.org/documentation

05

CICIDS2017 Dataset

Labeled benign and common attack traffic for IDS evaluation

unb.ca/cic/datasets/ids-2017.html

06

Production Validation Plan

Precision, recall, F1, false-positive rate, throughput, and latency

Project evaluation criteria

ACADEMIC NOTE

The interactive dashboard demonstrates the workflow; production claims require measured results from controlled experiments.